

This document compiles the instructions presented to participants during the experimental task. They are implemented in the experiment code (ptb\_demo.m, in the code/experiment folder of this repository) and are reproduced here as plain text for easier reading. The wording is identical to the code; only the formatting differs.

After the baseline recording, participants were presented with a general explanation. They advanced through the instructions at their own pace.

Alpha Waves: “When you are calm, relaxed and your mind quiet your alpha brain activity increases. When your mind is more active, such as when you are doing mental calculations, imaging scenarios or feeling stressed, your alpha activity becomes lower.”

Alpha Sonification: “In this experiment we will turn your alpha waves into sound. The louder the sound, the stronger your alpha waves. If you increase your alpha waves (keeping a calm mind), you will notice that the sound becomes louder. If you decrease your alpha waves (more busy mind), the volume of the sonification will go down.”

Artifacts: “Physical movements, such as jaw tension, head shifts, or opening your eyes during the experiment can heavily affect the calculation and distort the sonification. These changes do not reflect your true mental state. Try to remain still and keep your eyes closed.”

Exploration Phase: “Before we begin the training, you will listen to the sonification for one minute. The sonification is calibrated to your brain activity with eyes closed, so please keep your eyes closed during this phase. Focus on how you feel and how the sound responds. Keep in mind that the sound does not change instantly, there is a delay of about 4 seconds.”

Participants then completed a practice phase in which they rehearsed both the regulation and discrimination tasks under both feedback conditions before proceeding to the experimental phase. Participants were repeatedly reminded to keep their eyes closed, minimize movement, and follow the auditory instructions throughout the experiment.

“This experiment consists of two tasks: the regulation task and the discrimination task. You will repeat these two tasks several times, always in the same order: first regulation, then discrimination.”

Regulation Task: “Your goal is to manipulate your alpha waves. Initially, you will focus on increasing your alpha state by relaxing and maintaining a calm mind. If you reach the desired level, you will hear a short fanfare (Ta-da!). After a while, you will hear the voice instructing you to decrease your alpha state. You can try this by thinking about something stressful, remembering a tense moment, or imagining physical activity. You will also be informed whether you will hear the sonification or not during the upcoming task.”

Discrimination Task: "Instead of manipulating your alpha state, you will need to identify your current alpha state. During this phase you won't hear any sonification. But, every so often, you'll hear a bell sound. At that point, you will need to decide whether you were, just before, in a high alpha state (↑) or a low alpha state (↓), by pressing ↑ or ↓ on the keyboard. After your answer, you'll hear the short fanfare (Ta-da!) if your answer was correct."